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Review

Mitochondrial-derived vesicles in metabolism, disease, and aging

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SUMMARY

Mitochondria are central hubs of cellular metabolism and are tightly connected to signaling pathways. The dynamic plasticity of mitochondria to fuse, divide, and contact other organelles to flux metabolites is central to their function. To ensure *bona fide* functionality and signaling interconnectivity, diverse molecular mechanisms evolved. An ancient and long-overlooked mechanism is the generation of mitochondrial-derived vesicles (MDVs) that shuttle selected mitochondrial cargoes to target organelles. Just recently, we gained significant insight into the mechanisms and functions of MDV transport, ranging from their role in mitochondrial quality control to immune signaling, thus demonstrating unexpected and diverse physiological aspects of MDV transport. This review highlights the origin of MDVs, their biogenesis, and their cargo selection, with a specific focus on the contribution of MDV transport to signaling across cell and organ barriers. Additionally, the implications of MDVs in peroxisome biogenesis, neurodegeneration, metabolism, aging, and cancer are discussed.

MDVs AND THEIR ORIGIN

Mitochondrial-derived vesicle (MDV) trafficking first came to attention in 2008 with the discovery of a vesicular transport pathway transporting protein cargoes to peroxisomes.[1] A few years later, another vesicular trafficking route was discovered, directly connecting mitochondria to the endocytic pathway,[2] unequivocally integrating mitochondria in the concept of membrane trafficking. The capacity of mitochondria to shed vesicles is conserved in evolution since bacteria and archaea frequently generate vesicles.[3] Bacterial vesicles (BVs) are of pivotal physiological importance for diverse processes including bacterial communication, promoting pathogenesis, and bacterial survival.[4–6] Therefore, MDV transport pathways are consistent with its ancient evolutionary heritage as a free-living alpha-proteobacteria.[7]

The early experiments studying MDV transport posited three main criteria to uniquely define them. First, immuno-fluorescence microscopy visualized MDVs as small punctate structures with the distinct feature of cargo specificity. This was defined as a specific enrichment of a subset of mitochondrial proteins, whereas other mitochondrial proteins are excluded, pointing to an active cargo selection mechanism.[1,2,8–10] The characteristic of cargo specificity is a key determinant to differentiate MDVs from very small mitochondrial fragments generated by mitochondrial division. Second, electron microscopy (EM) analysis and immuno-gold staining confirmed the presence of biological membranes as a structural part of these round vesicular structure with a defined size of 60–150 nm.[1,2,9] Notably, EM analysis showed two different types of MDVs, namely single-membraned MDVs exclusively formed by the outer mitochondrial membrane and double-membraned MDVs containing both outer and inner mitochondrial membrane contents.[9] Third, MDV biogenesis did not require the autophagy machinery, thus separating MDVs from selective mitophagy pathways.[2,8] These main criteria help to define and differentiate MDVs from other selective pathways reshaping and degrading mitochondrial content.

THE PRINCIPLES OF MDV FORMATION

Currently, two mechanistic models have been described to generate MDVs: in the first model, electron-dense budding structures form on mitochondria and are subsequently released reminiscent of classical vesicular pathways, such as clathrin-mediated endocytosis.[1] In the second model, thin and relatively long membrane protrusions are pulled out of mitochondria along microtubules following by a scission event close to the tip of the protrusion,[11] thus resembling parallels to the process of Dynamin2-dependent autophagic breakdown of lipid droplets.[12] This second model for MDV formation was also observed in *Arabidopsis thaliana*[13] and in a high-throughput micrograph study of the brain in *Dugesia tigrine* (Planaria), *Mus musculus*, and *Drosophila melanogaster*.[14] Of note, these two models are not mutually exclusive but could reflect variations of the same mechanistic process utilizing overlapping molecular machineries, a mechanistic question that still awaits a final answer. Nearly 40 years of work in the secretory and endocytic pathways provide us with the fundamental principles that govern the formation of cargo-selected membrane vesicles.[15,16] Although our mechanistic understanding of MDV formation remains rudimentary, the guiding principles hold as we uncover the machineries and map them within the steps common to all vesicle transport pathways. These include the initiation phase, where cargoes are selected and vesicle formation is initiated, the tubulation phase,

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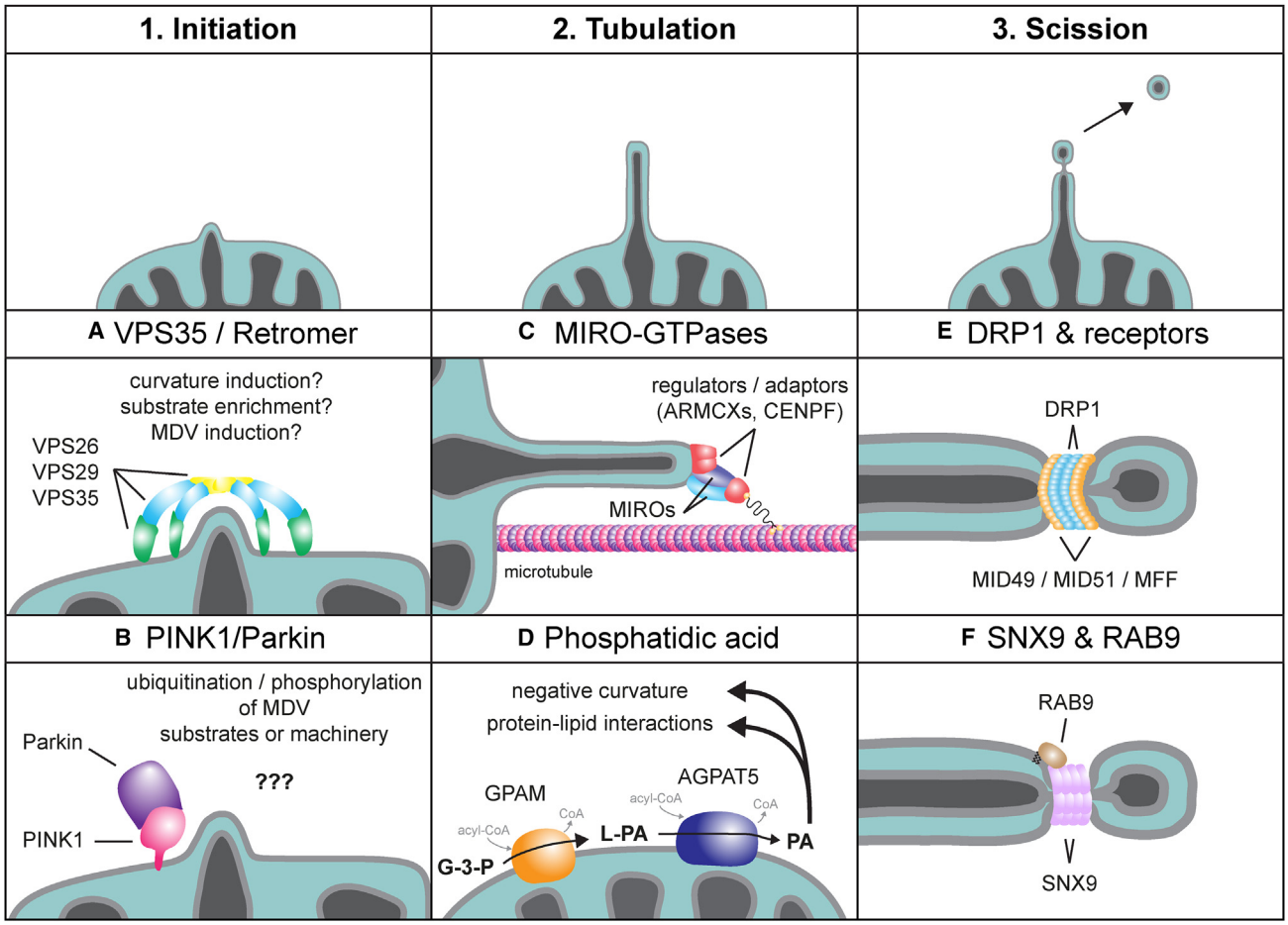


Figure 1. Overview of the MDV biogenesis process

Biogenesis of MDVs starts with an initiation event inducing initial curvature followed by a tubulation and membrane remodeling phase and is finally completed by membrane scission (steps 1–3). Several molecular players have been identified to catalyze these events based on the analysis of different subclasses of MDVs.

where we observe the extension or protrusion of the cargoes, and the final scission event that separates the vesicle from the mitochondrion (Figure 1). Once MDVs are formed, they are transported to the target organelle, for which we have much left to learn (Figure 2). For some classes of MDVs, the fusion with the late endosome requires canonical vesicle fusion machineries called soluble NSF-attachment receptors (SNAREs), but others appear to target the endomembrane or peroxisomal compartments using SNARE-independent mechanisms. Although there is a great deal of work still to be done, significant advances have been made in the last few years, as discussed below (Figure 1; Table 1).

PROTEIN MACHINERIES REQUIRED FOR MDV INITIATION

The first protein linked to MDV formation was the Parkinson's disease (PD)-related protein VPS35.[10] VPS35 acts in a trimeric complex together with VPS26 and VPS29, comprising the retromer retrieval complex known to play key roles in protein sorting back to the Golgi complex or to the plasma membrane.[25] The retromer complex binds cargo proteins through exposed amino acid motifs and enriches these cargoes into tubular structures along with the help of sorting nexin (SNX) proteins that induce membrane curvature.[26] VPS35 was biochemically identified as an interacting partner of the outer mitochondrial membrane protein mitochondrial anchored protein ligase (MAPL)/MUL1.[10,27] Live-cell microscopic analysis revealed that VPS35 is recruited to budding MDV structures and is requisite for the delivery of MAPL from mitochondria to peroxisomes.[10] This suggested that the cytosolic tail domain of MAPL was recognized by the retromer complex to drive vesicle transport to peroxisomes. Additional studies have shown effects on mitochondrial morphology and function in cells and neurons lacking Vps35 or knockin systems carrying the autosomal dominant PD-related mutation Vps35 D620N.[28–30] These studies demonstrated that VPS35 is required for the degradation of oligomeric dynamin-related protein 1 (DRP1) complexes through MDV-dependent delivery to lysosomes.[17,18] Although these studies provide evidence that VPS35 is required for the sorting of other cargoes into MDVs, the mechanism of recruitment and recognition of specific mitochondrial cargo tails has not yet been established. Presumably, the principles of retromer function seen at the endosome are similar to those at mitochondria, with the potential distinction

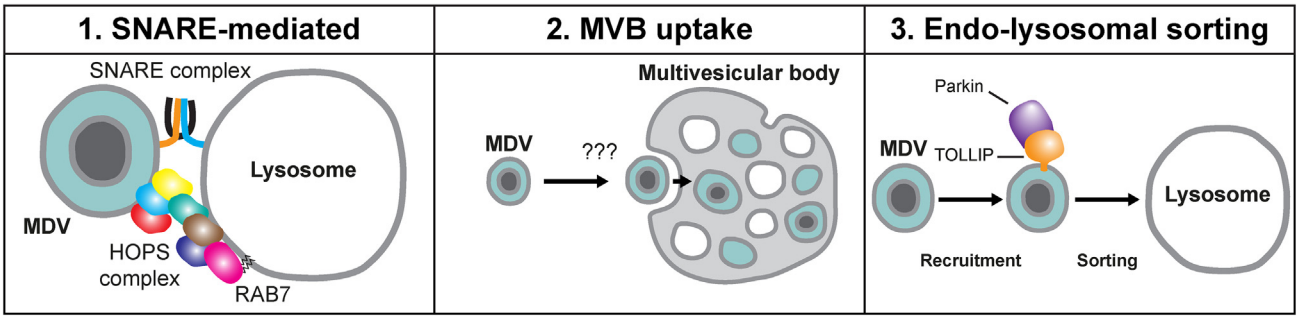


Figure 2. Overview of mechanisms of MDV clearance

MDVs can be cleared within the cell at least through three major mechanisms. First, MDVs can undergo SNARE-mediated fusion with the endolysosomal compartment. Second, MDVs can be delivered to the multivesicular body. The mechanistic details of this delivery are not resolved yet. Third, MDVs can be sorted into the endolysosomal pathway through the recruitment and action of Parkin and TOLLIP.

being in the mechanisms of signaled recruitment during specific stresses or other pathways.

Other PD-related proteins including the kinase PINK1 and the ubiquitin ligase Parkin were also linked to the initial biogenesis of MDV induced by mitochondrial stress in cultured cell lines.[2,8] The incorporation of different cargoes depends on the nature of the stress,[9] and a subset of these stress-induced MDVs required PINK1/Parkin for their formation.[8] PINK1 serves as a mitochondrial sensor and arrests in the import channel upon mitochondrial dysfunction.[31] Upon arrest, PINK1 phosphorylates the ubiquitin ligase Parkin as well as ubiquitin, which triggers a feedforward loop of ubiquitination and phosphorylation ultimately leading to mitophagy.[32,33] Kinetic analysis using a low dose of the mitochondrial complex III inhibitor antimycin A, which is commonly used at higher concentrations to acutely induce mitophagy, showed a burst of MDVs dependent on PINK1/Parkin, whose delivery to the late endosome was independent of the autophagy machinery.[8] In these experiments, mitophagy was only observed after 12–24 h of antimycin A treatment, whereas the peak of MDV formation was observed after 1–3 h. Similar evidence for MDVs as an early response to stress was seen upon addition of cannabinoids, where MDV formation preceded mitophagy and was dependent on PINK1 and Parkin.[34] Low-dose carbon monoxide treatment of oligodendrocyte precursor cells led to the generation of MDVs that functioned as a preconditioning mechanism to protect the cell from carbon monoxide toxicity.[35] Together this suggests that MDVs act early to sculpt the mitochondrial proteome by removing damaged or oxidized contents.[8,9,36]

Treatment of macrophages with lipopolysaccharide (LPS) or the non-phagocytosed *Citrobacter rodentium* was also shown to signal the generation of MDVs carrying the mitochondrial matrix enzyme 2-oxoglutarate dehydrogenase (OGDH), which were delivered to the late endosome/lysosome.[19,24] However, these MDVs activated by LPS-mediated Toll-like receptor 4 (TLR4) engagement did not require PINK1 and Parkin, and instead, these PD proteins repressed the formation of this class of MDVs. LPS activation led to the mitochondrial recruitment of sorting nexin 9 (SNX9), which was critical for the generation of inflammation-induced MDVs.[19] SNX9 is unique among SNXs since it has an extended amino terminus containing an SH3 domain; however, it also contains the PX-BAR (bin/amphiphysin/rvs) domain common to all SNX proteins and plays critical roles in membrane tubulation and vesicle formation.[37] It has not been shown to bind and function with the retromer complex described above; however, this cannot yet be excluded for MDVs. The presence of Parkin led to the proteasome-dependent degradation of SNX9, suggesting that SNX9 might be a ubiquitin substrate of Parkin upon LPS treatment, thereby potentially blocking MDV formation.[19] Additionally, the small guanosine triphosphatase (GTPase) RAB9 was recruited to mitochondria and shown to be required for the generation of inflammation-induced MDVs.[19] RAB9 canonically functions at the late endosome in regulating retrieval and recycling of proteins back to the Golgi and endosome compartments, as well as regulating autophagosome formation.[38] As with the VPS35 retromer complex, the mechanisms that govern RAB9 recruitment to mitochondria are unclear. RAB GTPase recruitment to membranes is driven by the presence and activation of the cognate GTP exchange factor (GEF), hinting that a mitochondrial RAB9 GEF must be recruited to mitochondria upon inflammatory stimuli. Of note, some other classes of MDVs, such as TOMM20-containing MDVs, are not regulated by PINK1/Parkin or SNX9,[11] again highlighting the plasticity and specificity of the molecular machinery involved in MDV formation.

COUPLING MDV FORMATION TO MICROTUBULE MACHINERIES

Although multiple PD-related proteins play modulatory roles in MDV formation,[8,10,18,19] the core machinery and mechanisms of vesicle budding had remained elusive for over a decade. A recent study shed new light on this process through the analysis of steady-state MDVs carrying the outer mitochondrial membrane (OMM) import receptor TOMM20.[11] A major finding of this work was the identification of a key role for microtubule-mediated pulling of thin membrane tubules to separate the selected cargo from mitochondria (Figure 1). Several microtubule-related proteins were enriched in isolated TOMM20-containing MDVs including MIRO1, MIRO2 (mitochondrial Rho GTPases), centromere protein F (CENP-F), and a number of armadillo repeat-containing proteins on the X chromosome (ARMCX) proteins.[11] The MIRO proteins are very well-characterized calcium-regulated OMM GTPases that link mitochondria to

Table 1. Key players involved in MDV biogenesis

Players	Molecular function	MDV process	MDV references
Cargo selection			
VPS26A	coat protein	cargo selection	Braschi et al.[10]
VPS35	coat protein	cargo selection	Braschi et al.[10] and Wang et al.[17,18]
Parkin	E3 ubiquitin ligase	post-translational modifications	McLelland et al.,[8] Matheoud et al.,[19] Todkar et al.,[20] Roberts et al.,[21] Abuaita et al.,[22] and Towers et al.[23]
PINK1	kinase	post-translational modifications	McLelland et al.,[8] Matheoud et al.,[19] Roberts et al.[21] and Matheoud et al.[24]
Membrane rearrangement and vesicle initiation			
SNX9	sorting nexin	tubulation/constriction	Matheoud et al.,[19] Todkar et al.,[20] and Towers et al.[23]
RAB9	small GTPase	recruitment of effectors?	Matheoud et al.[19]
Interaction with the cytoskeleton			
MIRO1	mitochondrial RHO GTPase	membrane tubulation	König et al.[11]
MIRO2	mitochondrial RHO GTPase	membrane tubulation	König et al.[11]
CENP-F	MIRO interactor	unknown	König et al.[11]
ARMCX1	MIRO interactor	unknown	König et al.[11]
ARMCX3	MIRO interactor	unknown	König et al.[11]
Membrane scission			
DRP1	dynamin-related GTPase	membrane scission	König et al.[11]
MFF	DRP1 receptor	DRP1 recruitment	König et al.[11]
MID49	DRP1 receptor	DRP1 recruitment	König et al.[11]
MID51	DRP1 receptor	DRP1 recruitment	König et al.[11]
OPA1	dynamin-related GTPase	unknown	Todkar et al [20]
Alteration of phospholipid composition			
GPAM	L-PA synthesis	curvature induction/DRP1 regulation?	König et al.[11]
AGPAT5	PA synthesis	curvature induction/DRP1 regulation?	König et al.[11]

the kinesin/dynein family of motor proteins.[39] CENP-F was shown to couple mitochondria to the growing ends of microtubules during mitosis through interactions with MIRO1,[40,41] and the understudied ARMCX proteins are highly expressed in neurons where at least one member has been shown to regulate mitochondrial motility in neurons through complexes with MIRO1.[42,43] Loss of these proteins inhibited vesicle formation, consistent with an essential role for microtubule-mediated extraction of cargo-selected mitochondrial tubules, which was further demonstrated by live-cell microscopy.[11] Similar images and videos of the striking tubulation that precedes MDV formation were documented using the outer mitochondrial membrane marker OMP25-GFP with high-resolution SIM microscopy.[44] Additionally, a role of the fusion GTPase OPA1 was suggested for the incorporation of matrix and inner membrane cargoes into MDVs.[20]

A detailed proteomic and lipidomic analysis of TOMM20-containing MDVs revealed an enrichment of phosphatidic acid in MDVs, along with at least 2 enzymes (glycerol-3-phosphate acyltransferase 1 [GPAM] and 1-acyl-sn-glycerol-3-phosphate acyltransferase epsilon [AGPAT5]) that catalyze the *de novo* biogenesis of phosphatidic acid on mitochondria.[11] Depletion of phosphatidic acid from the mitochondrial OMM blocked MDV formation, indicating that active lipid remodeling is important.[11] The constitutive generation of TOMM20-containing MDVs did not require Parkin or SNX9, consistent with the evidence that these are modulators of stress-induced MDVs but not part of the core mechanisms of vesicle formation.[11] The list of ~100 proteins in the proteomic analysis almost certainly holds more clues to help define these mechanisms.[11]

MECHANISMS OF MDV SCISSION

Initially, small interfering RNA (siRNA)-mediated knockdown approaches also suggested that MDV biogenesis did not require the mitochondrial fission machinery[1,2]; however, recent work using a CRISPR-Cas9 knockout approach demonstrated that the fission machinery is essential for the final scission step in MDV biogenesis.[11] Live imaging further revealed that the fission GTPase DRP1 is recruited near the tip of an extending tubule. The DRP1 foci at MDV scission sites were considerably smaller in size from mitochondrial fission events and more dynamic. Although these qualitative differences suggest kinetic and mechanistic distinctions, MDV scission required the same DRP1 adaptor proteins MFF, MID49, and MID51.[11] The regulation of DRP1 recruitment to mitochondrial division sites is tightly linked to cellular signaling pathways that lead to phosphorylation of DRP1 and MFF, along with other post-translational modifications.[45] Interestingly, MDV formation was differentially regulated in a number of conditions and was not modulated by the

Table 2. Key players involved in MDV delivery to target organelles

Players	Molecular function	MDV process	MDV references
Fusion with the lysosome			
STX17	SNARE protein	membrane fusion	McLelland et al.[46]
SNAP29	SNARE protein	membrane fusion	McLelland et al.[46]
VAMP7	SNARE protein	membrane fusion	McLelland et al.[46]
VPS39	HOPS complex subunit	tethering	McLelland et al.[46]
VPS41	HOPS complex subunit	tethering	McLelland et al.[46]
RAB7	small GTPase	recruitment of effectors	McLelland et al.[46]
Endolysosomal sorting			
TOLLIP	adaptor protein	unknown	Ryan et al.[47]
Parkin	E3 ubiquitin ligase	unknown	Ryan et al.[47]

established phosphorylation states of DRP1.[11] Therefore, although MDVs require DRP1 for the final scission event, the regulation of DRP1 recruitment to these sites remains unclear. Whether or not the endoplasmic reticulum (ER) or other organelle contact sites may contribute to MDV scission remains to be determined.

TARGETING THE ENDOLYSOSOMES

Once the MDV is formed, it must target its final destination (Figure 2). Therefore, besides elucidating the core mechanisms of MDV formation, identifying the molecular details of fusion of MDVs with their target organelle is equally important (Table 2). The characterization of the TOMM20-containing MDV proteome did not identify any canonical SNAREs, leaving the mechanism of fusion with the target organelle still elusive for this class of MDVs.[11] It is possible that this class of MDVs may be internalized into the multivesicular body through a microautophagy pathway. In contrast to TOMM20-containing MDVs, the delivery of stress-induced PINK1/Parkin-mediated MDVs to the late endosome required the SNARE protein syntaxin 17 (STX17).[46] STX17 forms a SNARE complex with SNAP29 and VAMP7 to allow the fusion of MDVs with the late endosome/lysosome.[46] Loss of STX17 led to the cytosolic accumulation of Parkin-YFP-labeled MDVs as visualized by both confocal and immunogold EM.[46] Additionally, the small GTPase RAB7, which regulates endolysosomal fusion processes, and VPS39/VPS41, two subunits of the HOPS tethering complex, have been shown to act together with STX17 in the delivery of stress-induced PINK1/Parkin-dependent MDVs to the late endosome/lysosome.[46]

Parkin was also suggested to control the targeting of MDVs into the endolysosomal compartment together with its multifunctional partner, the Toll-interacting protein (Tollip).[47] Like Parkin, Tollip is linked to immune signaling and was shown to inhibit TLR activation,[48,49] to stabilize the cytosolic DNA sensing protein STING in the ER,[50] and to play roles in ESCRT-mediated protein sorting through its interaction with the ESCRT-0 subunit TOM1.[51–53] During oxidative stress, Tollip was observed on one class of MDVs (TOMM20+/PDH−) that were also positive for Parkin.[47] Tollip interacts with the UBL domain of Parkin and ubiquitin, which is required for its MDV association.[47] Unlike for the generation of stress-induced MDVs,[8] Parkin's ubiquitin ligase activity was not required for the targeting of MDVs into the endolysosomal compartment.[47] Of note, Parkin is not essential to target all classes of MDVs into the endolysosomal compartment.[8,11] One possible explanation for these differences may be due to the immune functions of Tollip, where ectopically expressing or silencing of Tollip may itself drive signaling events that initiate MDV formation, cargo selection, and transport. What is becoming clear is that the fine-tuning of MDV transport allows the tight regulation of these pathways with the signaling state of the cell.

We have highlighted progress in dissecting the mechanisms that underlie MDV formation; however, new challenges emerge. Most critical is that all of the protein machineries identified to date have additional functions in the cell. This confounds the interpretation of any loss-of-function experiment since the result may be a loss of MDVs, but also a loss in, for example, mitochondrial motility (MIRO1/2), mitochondrial and peroxisomal fission (DRP1), or perhaps indirect effects of alterations in endocytosis (SNX9) and the endomembrane system (VPS35). Until MDV-specific protein machineries are identified these challenges will remain.

PLASTICITY IN MDV CARGO SELECTION

Early on, cell-free reconstitution reactions from isolated mitochondria revealed a flexibility in cargo incorporation, where distinct toxins would promote inclusion of different mitochondrial content enriched in oxidized proteins.[9] A pattern emerged that the matrix contents included subunits of large, megadalton complexes, such as pyruvate dehydrogenase, OGDH, and some complexes of the electron transport chain.[8,9] Based on these early observations, two proteomic studies have adopted the cell-free reconstitution reaction to map the proteome of MDVs in the context of oxidative stress using cardiac or neuronal mitochondria isolated from the heart or brain, respectively.[21,54] These proteomic studies confirmed initial observations and supported the idea that MDVs extract large complexes that may be difficult (and/or dangerous) to disassemble and degrade within the organelle. However, the assembly state of these complexes remained unclear. The recent analysis of immuno-isolated TOMM20-containing MDVs using blue-native PAGE demonstrated that electron transport chain complexes or the mitochondrial protein import machinery remained fully intact, indicating that there was no proteolytic processing or disassembly of these large complexes prior to incorporation into MDVs.[11] Together, these studies have revealed a strong flexibility in cargo

composition based on the subclasses of MDVs, the isolated tissues, and the applied stresses, highlighting the vast complexity of MDV cargo in response to different triggers.[11,21,54]

Reminiscent of this complexity, proteomic studies of BVs have revealed a large number of cargoes from different compartments, such as the cytoplasm, periplasm, and bacterial inner or outer membranes, with a strong enrichment of β-barrel proteins.[3,5,55] This feature is also conserved in the proteome of TOMM20-containing MDVs, in which all mitochondrial OMM β-barrel proteins (VDAC1-3, TOMM40, and SAMM50) were identified.[11] Remarkably, β-barrel proteins are fully membrane-embedded and oligomerized in MDVs and are classified as intrinsically difficult to extract proteins.[11]

What is very poorly understood is how these cargoes and oligomeric protein complexes are chosen for incorporation into MDVs. To date, TOMM20-containing MDVs are the most abundant class of MDVs generated in steady state.[2,11] It is possible that this reflects stalled import complexes that cannot be cleared by the mitochondrial compromised protein import response (mitoCPR) and are therefore removed in MDVs.[56] Another possible mechanism contributing to cargo selectivity might be the process of tubulation itself, as the increased surface-to-volume ratio would favor membrane-bound complexes over soluble matrix components. Whether or not the MIRO1/2 machinery may link the import machinery to other cargoes and the mechanisms to retain inner membrane contacts to generate double-membraned vesicles is also unclear. MIRO1 was shown to form a complex with the mitochondrial intermembrane space bridging complex (MIB), containing both MICOS and SAMM50.[57] SAMM50 is a component of the import machinery that facilitates the insertion of transmembrane domains into the bilayer.[58,59] At least a pool of SAMM50 is also integrated within the MICOS complex that stabilizes contact sites between the outer and inner membrane.[60,61] Therefore, the potential for these proteins as regulatory nodes that direct the generation of single- or double-membraned vesicles should be investigated.

In contrast to steady-state MDVs, many of the stress-induced MDVs do not contain the import complexes. Therefore, their mechanisms of cargo selection must be distinct. Oxidative stress-induced MDVs are often modulated by the PD-related factors discussed previously.[8,19] It is our hypothesis that MDV transport is activated in response to metabolic and cell fate transitions, which means that the structural features that drive cargo selection must be intimately coupled to the metabolic state. For example, cargoes may change confirmation upon the direct modification by a given metabolic substrate. This could be reflected in numerous ways, from ROS-mediated oxidation, glutathionylated, conjugation by reactive TCA cycle intermediates succinylation or itaconylation, or possibly through more direct signaling events involving phosphorylation or NAD^+-dependent deacetylation by sirtuins.[62] We envision that the modification of protein complexes by these metabolites could lead to their aggregation or drive interactions with machineries coupled to MDV formation. Aggregated or even phase-separated cargoes could potentially initiate membrane curvature from the matrix side, which in turn would recruit cytosolic proteins like SNXs, to initiate vesicle or tubule formation from the cytosolic side. This would allow an intrinsic flexibility for mitochondria to release any large, "failing" protein complex within vesicles and would provide plasticity depending on the tissue and specific stressors. In the case of inflammation, there is a spike in ROS and itaconate (metabolic remodeling) that could lead to additional modifications to drive susceptible complexes like OGDH into vesicles.[63,64]

Exciting recent developments have documented a new class of MDVs that form in response to disease mutations in fumarate hydratase, leading to an accumulation of fumarate.[65] The elevation in fumarate led to the incorporation of mtDNA into MDVs that required SNX9 for their formation. Once released from mitochondria, the mtDNA within MDVs gained access to the cytosol where it activated cGAS and STING pathways that triggered innate immune signaling.[65] This indicated that the accumulated metabolite somehow led to the specific incorporation of mtDNA into MDVs, perhaps through covalent modifications on transcription factor A, mitochondrial (TFAM), that packages mtDNA into the nucleoid structures. There is an emerging appreciation of TCA cycle intermediates, particularly succinate, as a post-translational modifier through an acylation reaction on acceptor lysine residues.[66] This acylation reactions require "writers, readers, and eraser" proteins that drive the conjugation, interact with modified substrate, and reverse the conjugation. We envision that these reactions will directly couple metabolism to the generation of MDVs. For now, how cargoes are incorporated into MDVs remains one of the most critical unanswered questions in the field.

FATES OF MDVs

The targets of MDVs include peroxisomes, late endosomes, phagosome, and lysosome (Figure 3). The first discovered MDV transport pathway described the delivery of the outer mitochondrial membrane protein MAPL to peroxisomes.[1,10] Although MAPL was transported to a subset of peroxisomes in steady state, it was also shown that the birth of new peroxisomes requires a vesicle transport pathway from mitochondria.[67] Fibroblasts from Zellweger patients lacking the core peroxin PEX3 are completely devoid of peroxisomes. Upon rescue with the missing PEX3 protein, new peroxisomes are born.[68] It had long been known that PEX3 initially targeted mitochondria,[69] whereas its partner in peroxisomal protein import, PEX16, initially targeted the ER in cells lacking peroxisomes.[70] By following the birth of new peroxisomes, it was shown that PEX3 and PEX16 were each sorted into vesicles from their respective organelles.[67] Only upon fusion of the two pre-peroxisome populations could the import complex be assembled, allowing the development of a mature peroxisome.[67] Thus, MDV-mediated cargo delivery to peroxisomes plays an essential part in the *de novo* biogenesis of peroxisomes.[67] Current evidence suggests that some peroxisomal transmembrane proteins may utilize either the ER SEC61 translocon (in the case of PEX16),[70,71] or the mitochondrial TOM translocon (i.e., MAPL or PEX3)[1,67,69] for initial biogenesis, rather than inserting through a peroxisomal import translocon. Recent structural studies have demonstrated how the membrane-embedded peroxisomal ubiquitin ligase import complex of PEX2/10/12 forms a pore structure with similarities to the SEC61 or TOM machineries in ER and mitochondria.[72] However, it is not clear yet how complex, multispanning membrane proteins are inserted laterally into the bilayer through this kind of pore and how this may be structurally facilitated by peroxisomal membrane protein import subunits PEX3/16/19. We consider

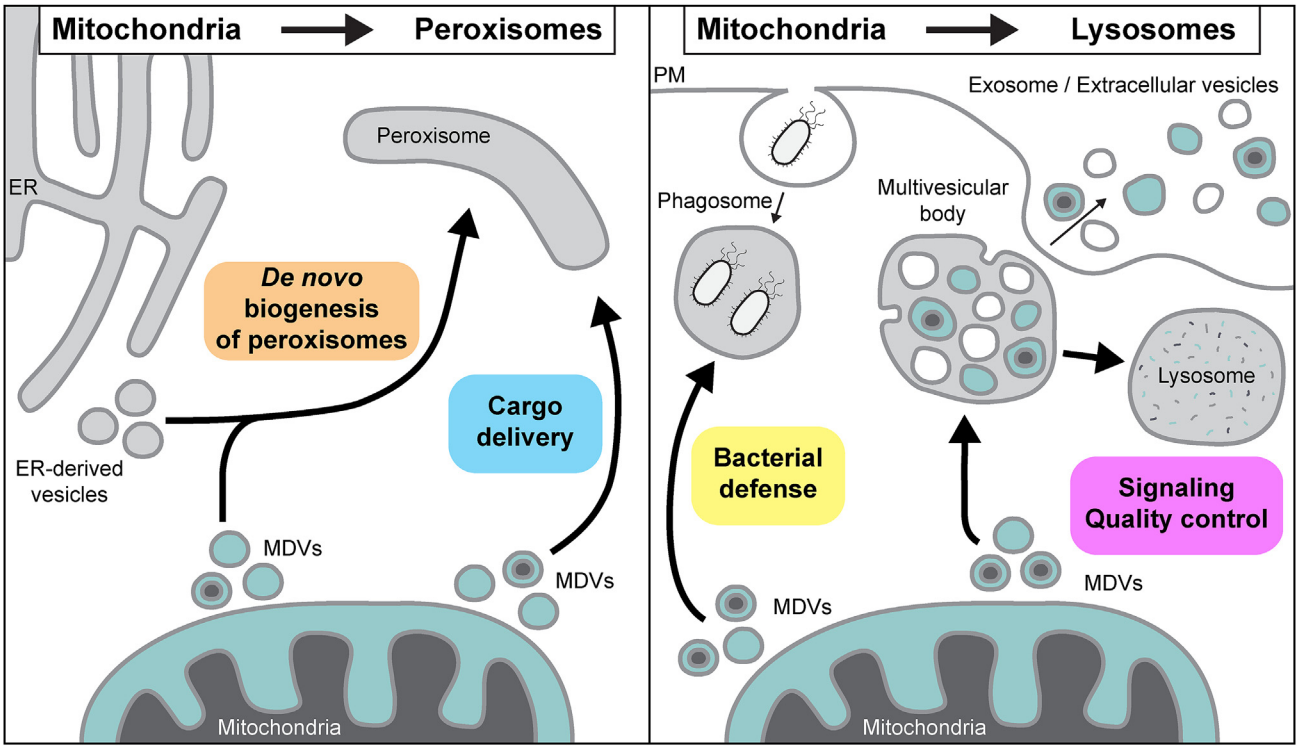


Figure 3. Different fates of MDVs

Overview of the different fates of MDVs discovered today. MDVs can deliver protein cargoes to peroxisomes or contribute to the *de novo* biogenesis of peroxisomes by fusing with specific ER-derived vesicles containing peroxisomal proteins (left). The major transport route delivers MDVs from mitochondria to the endolysosomal system (right). Within the endolysosomal pathway, MDVs can modulate the bacterial defense of the cell and contribute to quality control through lysosomal degradation as well as signaling pathways for example by release of mitochondrial content through exosomes or extracellular vesicles.

there are dual pathways for peroxisomal import of membrane-anchored proteins, either directly into the organelle through the PEX2/10/12 translocon or by delivery through MDVs and ER-derived vesicles.

The fate of mitochondrial content within the late endosome/lysosome is primarily degradation as part of the continuum of quality control pathways.[73] Mitochondrial proteases in the matrix and intermembrane space offer the first line of quality control, degrading oxidized and/or misfolded proteins.[74] In addition to mitochondrial proteases, many OMM proteins are degraded in a ubiquitin-mediated, retrotranslocation into cytosol for degradation within cytosolic proteasomes.[75] MDVs are at the next step of the quality control continuum, removing larger protein complexes and membrane proteins that may be refractive to proteolytic degradation within the mitochondria. Although we refer to these levels of quality control as a continuum, they will occur simultaneously, depending on the client portfolio of each pathway. Lastly, when entire organelles are dysfunctional, they can be targeted and removed by mitophagy.[76]

In addition to the quality control function of MDVs, recent studies revealed more active signaling and metabolic functions of MDV cargoes. The fact that many protein complexes are fully intact in MDVs offers the interesting possibility that, at least in some conditions, these MDVs could be an active source of metabolites.[11] Recent work in yeast demonstrated that MDVs can carry a functional ATP synthase complex that generates ATP and retains electrochemical potential.[77] These MDVs were seen to fuse with naive mitochondria rather than targeting the late endosome or vacuole. Another example is the MDV-dependent delivery of SOD2 to an MRSA-containing autophagosome, which aids in killing the pathogen.[22] Further supporting a role of MDVs in immunity and inflammatory signaling, MDV cargoes that are expelled from the cell can serve as signals to trigger responses across cell types or even tissues.[20,78] Therefore, MDVs first target the multivesicular body from where they can be shuffled to the extracellular space through exosomes and extracellular vesicles (EVs).[20] Altogether, MDVs can target various compartments within the cell to serve diverse functions from basic quality control to complex signaling cascades.

ALTERNATIVE MITOCHONDRIAL REMODELING PATHWAYS

Besides MDVs, alternative pathways exist to specially remodel mitochondria, including mitophagy (Figure 4). Although the relatively small size is a key feature of MDVs, there are at least two very exciting examples where mitochondria can generate very large, cargo-selected compartments *de novo*. Work in yeast has shown that conditions of high amino acids induce the generation of a very large, micron-sized structure, termed mitochondrial-derived compartment (MDC) enriched in metabolite transporters.[79] This MDC is maintained for hours and is a metabolically active compartment that converts the excess amino acids to fusel alcohol before ultimately being degraded.[80]

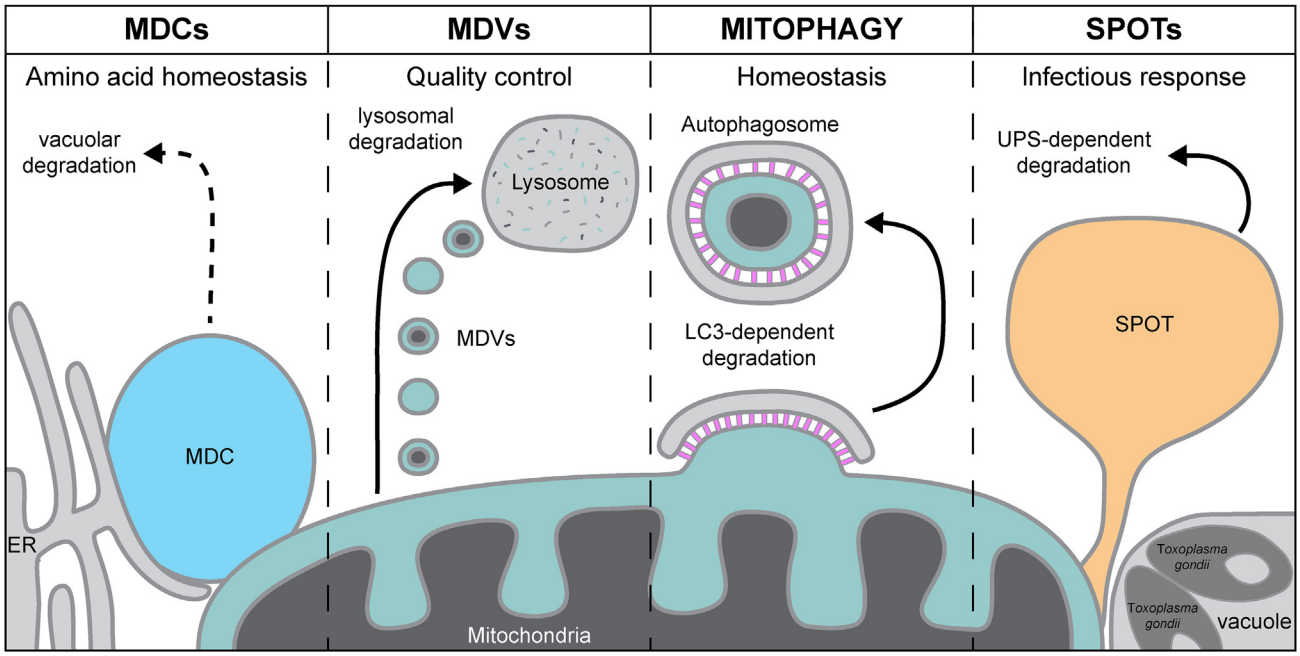


Figure 4. Overview of various mitochondrial membrane remodeling pathways

Several mechanisms have been described to act on the outer mitochondrial membrane to reshape the mitochondrial proteome. In yeast, mitochondrial-derived compartments (MDCs) can form a functional, metabolically active compartment in response to amino acid stress and can modulate the degradation of a subset of the mitochondrial proteome. Mitochondrial-derived vesicles (MDVs) can selectively deliver mitochondrial contents for lysosomal degradation at steady state or in response to various stress conditions, such as reactive oxygen species. The most studied remodeling pathway is LC3-dependent degradation, for example, by piecemeal or bit-by-bit mitophagy. Additionally, *Toxoplasma* infection can trigger the formation of large mitochondrial structures named structures positive for OMM (SPOTs), which lead to the degradation of mitochondrial proteins through the ubiquitin-proteasome system (UPS).

Moreover, this demonstrated that mitochondria have retained the capacity to drive metabolic subcompartmentalization through the generation of cargo-selected MDCs, and even peroxisomes, with metabolic functionality. For the second example, recent work showed that parasitic trypanosome infection initiated the formation of large OMM compartments called structures positive for OMM (SPOTs).[81] It's unclear whether this dramatic event favors the host or the parasite, but the selective removal of OMM components led to import failure and presumably a number of other consequences.[81] This highlights the incredible plasticity of mitochondrial membranes to generate new structures *de novo* in highly variable biological systems and conditions.

A central feature of MDV transport is the independence from the autophagy machinery.[2] Consistent with this, a recent study has shown that MDVs can compensate in cells lacking core autophagy proteins like ATG5.[23] However, some have observed LC3 recruitment to Parkin-positive small mitochondrial fragments, termed piecemeal or bit-by-bit mitophagy.[36,82,83] Most studies do not examine these structures at the EM level, making it difficult to determine whether they are MDV related, or fragments of mitochondria undergoing canonical mitophagy. If the ubiquitin status on the surface of MDVs was high, it could lead to recognition by autophagy receptors and engulfment into an autophagosome.[32] There will be multiple levels of intersection between these pathways that will require further understanding of early events in the generation of these structures. Going forward, it is critical to use physiological conditions to interrogate the relationship between MDVs and mitophagy/autophagy. The use of proton uncouplers and other drugs that rapidly kill mitochondria has provided important insights into the mechanisms of acute mitophagy but has delayed progress in dissecting the complex events in conditions where mitochondria are compromised but remain active. This is most obvious in inherited mitochondrial disease patients, where organelles lacking entire electron transport chains, or loss of mtDNA, yet mitophagy is limited, displaying a frustrating lack of clearance of dysfunctional organelles.[84,85]

One critical question is the nature of the switch, or potential integration, between MDVs and mitophagy. Kinetically, low-dose antimycin A or cannabinoid treatment showed that MDVs precede mitophagy,[8,34] suggesting the cumulative damage must be broad enough for all import channels to accumulate PINK1 and amplify the ubiquitination reactions of Parkin to recruit autophagy adaptors. Importantly, multiple studies have shown that Parkin-dependent ubiquitination and proteasomal degradation of MIRO were required to arrest mitochondrial motility and allow mitophagy to proceed.[86,87] A secondary consequence of MIRO loss would be an arrest in MDV formation,[11] which may contribute to the switch to promote mitophagy. This sets a rheostat where the capacity to generate MDVs depends on the extent of dysfunction, potentially linked to the status of MIROs. Importantly, loss of PINK1 or Parkin in multiple models does not induce significant mitochondrial dysfunction, indicating that other mitophagy and proteostasis mechanisms can play critical roles in mitochondrial quality control.[88–90]

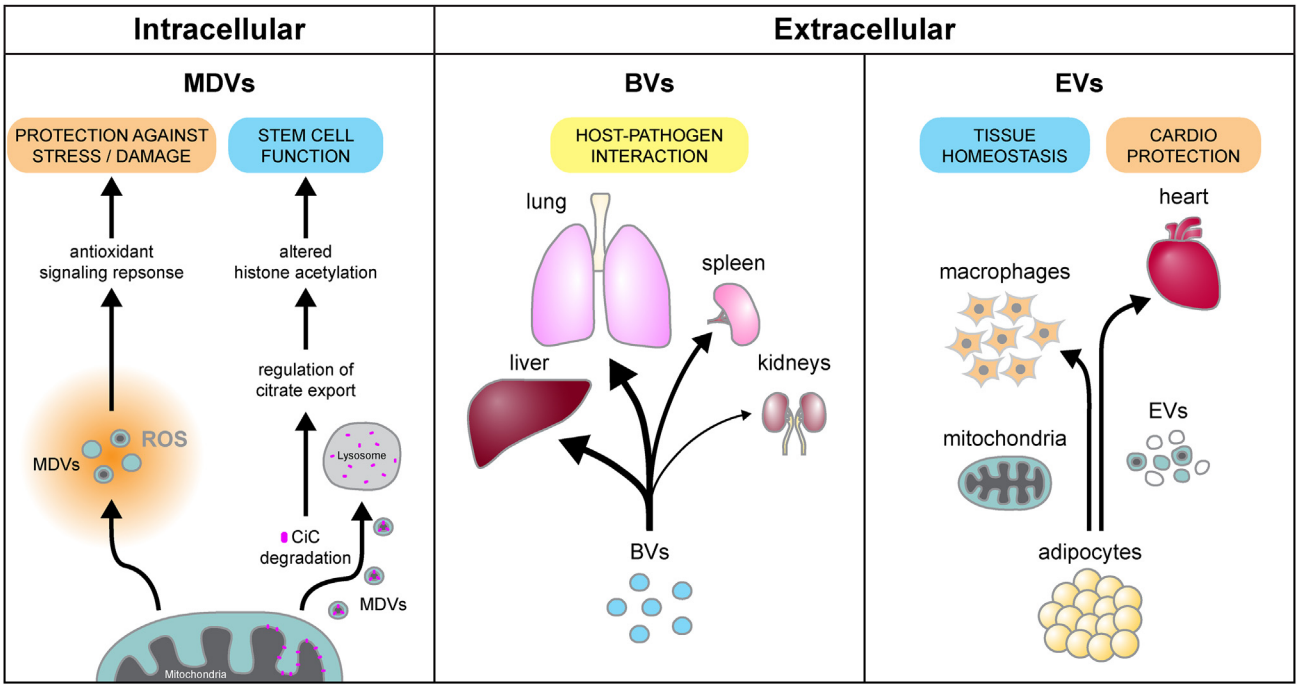


Figure 5. Metabolic implications of vesicular trafficking in and across cells

MDVs and bacterial vesicles (BVs) are implicated in several physiological processes. Intracellular, MDVs can play a protective role against various stresses and can impact epigenetics and thereby stem cell function. Reminiscent of BVs, which can travel large distances in the body through the bloodstream to stimulate host-pathogen interactions in various tissues and organs, extracellular vesicles (EVs) containing mitochondrial contents can regulate tissue homeostasis and prime target tissues like the heart to confer increased level of protection against stress.

Whether or not MIRO loss is central to all forms of mitophagy is not yet clear, so the transitions from MDV formation to full mitophagy require investigation in multiple contexts.

In conclusion, understanding the integration of the many mechanisms of quality control is essential in order to find ways to selectively enhance these mechanisms in disease models.

PHYSIOLOGICAL IMPACT OF MITOCHONDRIAL CONTENT IN EVs

One intriguing consequence of MDV delivery to the multivesicular body is the potential to secrete selective mitochondrial content from the cell. Recent studies indicate that MDVs can be directly captured within ectosomes budding from the plasma membrane or they may be released as part of the exosome pool when multivesicular bodies fuse with the cell surface.[20,78,91–93] Indeed, proteomic studies of EVs identified a large set of mitochondrial proteins with an emerging array of functional consequences.[94–96] These release mechanisms can play distinct roles in interorgan signaling, immune regulation, and even mitochondrial exchange.[93] The contributions of mitochondrial content transfer between cells in signaling pathways are an exciting and emerging area of research. There may be some evolutionary similarity to the function of BVs as drivers of host cell signaling pathways both within the microbiome and upon infection.[97] Several studies have identified BVs circulating in the bloodstream, thereby even reaching far-away tissues and target regions with emerging therapeutic relevance.[97,98] Thus, the concept of vesicular carriers as signaling units within and between cells, tissues, and even organisms is very well established in nature and is not restricted to MDVs (Figure 5).

Recent advances illustrating the functional importance of MDV release *in vivo* are being made in an expanding number of tissue contexts.[78,99] Given the technical challenges in studying MDVs and EVs *in vivo*, not all studies distinguish clearly whether the mitochondrial release reflects whole organelles or MDVs/exosome content. A common theme for MDV secretion is the clear role for specific mitochondrial stress, or inflammatory signaling, as the cue to release content. For example, inducible expression of mitochondrial ferritin in adipocytes allowed the specific induction of oxidative stress, particularly when mice are fed a high-fat diet.[78] Although the stress was activated within adipocytes, profound effects were seen in the heart, which were driven by adipose-derived EVs containing mitochondrial content. The authors demonstrated that the EVs were derived from exosome secretion, and although the content showed increased oxidation, the vesicles retained functional respiratory capacity. The recipient cardiomyocytes in the heart were seen to internalize the EVs, prompting an initial burst of ROS that led to tissue remodeling and protection against subsequent ischemia/reperfusion (I/R) injury. EVs derived from Parkin−/− adipocytes showed reduced mitochondrial protein content and did not provide protection in the heart when injected into wild-type mice experiencing I/R injury.[78] This provided critical new evidence for potentially long-range impact of secreted EVs carrying mitochondrial content as a messenger between adipocytes and the heart, particularly in conditions of obesity and adipocyte stress.

Secretion of mitochondria from multiple cell types has been seen as a mechanism for clearance by macrophages. A recent study revealed that mitochondria-containing EV release was stimulated from brown adipocytes during cold exposure.[99] Thermogenically induced isolated brown adipocytes led to increased mitochondrial uncoupling and ROS production that drove the formation of PINK1-dependent PDH+/TOMM20− MDVs that targeted the endolysosomal compartments prior to exosome secretion. The PDH content of the MDVs and EVs was shown to be cysteine oxidized, and treatment with n-acetyl cysteine blocked EV release. These EVs were cleared *in vivo* by patrolling monocyte-derived macrophages, a specific subtype to thermogenic conditions that showed reduced inflammation.[99] Loss of these macrophages led to accumulation of EVs and failure of the thermogenic response to cold exposure within the brown adipocytes. Therefore, the capacity of macrophages to clear EVs was critical for thermogenesis. Indeed, incubation of isolated brown adipocytes with EVs disrupted the typical thermogenic response including PPARγ signaling, decreased UCP1 expression, respiration, and altered TCA metabolites, at least in part through the activation of AMPK.[99]

Additionally, numerous groups have reported mitochondrial transfer between cells *in vivo*, a process mechanistically complex and poorly understood. Whole organelles can be transported in thin nanotunnels or within ectosomes that shed from the plasma membrane.[100–104] One example among many is the exchange of mitochondria from adipocytes to macrophages *in vivo*.[105] In an elegant study combining a fluorescent reporter mouse line with bone marrow transplantation, the transfer of mitochondria in white adipose tissue from adipocytes to neighboring macrophages was clearly demonstrated.[105] Importantly, these transferred mitochondria were not cleared by their new host but resulted in an increase of mitochondrial ROS production in these macrophages demonstrating metabolic activity.[105] Physiologically, the transfer of mitochondria from adipocytes to macrophages was shown to be crucial for the regulation of white adipocyte tissue homeostasis, which is impaired in obesity.[105]

Exosome release is also an established contributor to cellular migration through multiple mechanisms.[106] Recent studies have shed new light on the function of mitochondrial proteins and mtDNA released within exosomes in cancer models. Among the events leading to metabolic rewiring in cancer cells is the upregulation of a cystine/glutamate antiporter that effluxes glutamate in exchange for cystine, a critical step to generate the antioxidant glutathione required to counter the stress of the cancer cell.[92] This results in increased extracellular glutamate in the tumor environment, which can activate the glutamate receptor 3, promoting endosomal transport and exosome release that facilitates migration. However, chronic glutamate exposure also led to mitochondrial respiratory stress that increased mitochondrial content within the exosomes. Unlike the examples described above, this included mtDNA.[92] Although the study did not investigate the detailed mechanisms of content delivery to the late endosome compartments, the process was independent of autophagy machinery and required PINK1. Interestingly, PINK1, but not the kinase activity of PINK1, was requisite for mtDNA release in exosomes. This is somewhat reminiscent of studies with the Parkin partner Tollip described above, where the ubiquitin ligase activity of Parkin was not required for TOMM20 delivery to the multivesicular body/late endosome.[47] The authors suggested that glutamate stress through Glut3 signaling induced MDV transport to deliver mtDNA to the late endosome. The functional consequence of mtDNA within released EVs was to activate TLR9 receptors that further promoted endocytosis, migration, and invasiveness of the tumor.[92] Horizontal transfer of mtDNA within exosomes has been reported in other cancer studies, where it was shown to promote exit from dormancy of cancer stem-like cells through integration of the mtDNA into acceptor cells,[107] perhaps following mechanisms explored in the PINK1-dependent pathway above.[92]

The release of oxidized mtDNA from neutrophils was also described in LUPUS models, where again the mechanisms were not fully resolved but revealed an MDV pathway through the multivesicular body and release through exosomes.[91] These, along with more recent work[20,65] revealed that mtDNA nucleoids or DNA fragments can be MDV cargo given the right stimulus. Proteomics studies have identified mtDNA nucleoids in autophagosomes from brain and primary neurons, so these pathways can both contribute to the delivery of mitochondrial content to the endolysosomal and autophagosomal compartments.[108] Armed with rapidly emerging tools and mechanisms to selectively modulate MDV formation, future work will clarify the nature of mtDNA transport and release in EVs. There is a growing interest in the functional impact of mitochondrial content secreted from tissues in multiple disease paradigms. Although some will involve the transfer of intact whole mitochondria, the contribution of MDV content to the multivesicular body appears to contribute significantly to interorgan signaling, immunity, and cancer progression in multiple ways.

PATHOPHYSIOLOGICAL IMPACT OF MDV TRANSPORT

In addition to the consequences of MDV content release from cells, there is increasing insight into the roles of MDV transport in neurodegenerative disease, heart metabolism, and aging (Table 3).[8,19,24,81,109,110,111] Besides the removal of damaged protein complexes, MDV transport is a mechanism to selectively sculp the mitochondrial proteome to alter function, metabolism, and signaling. This is perhaps best illustrated in a study investigating mild oxidative damage in neurons.[109] The study investigated the response to very low-dose antimycin A (inhibitor of complex III, 1,000× lower than most studies) that did not elicit mitophagy or Parkin recruitment.[109] Instead, they observed a selective loss of the mitochondrial membrane protein Syntaphilin, which anchors mitochondria within the terminals. High-resolution imaging, video microscopy, and immunoelectron microscopy (immuno-EM) analysis revealed the release of Syntaphilin in MDVs that hitchhiked with endosomes back to the soma for degradation. The mitochondria within the terminals were then liberated from their anchored state, showing enhanced dynamics and retrograde transport.[109] The authors further examined this phenomenon in early asymptomatic models of fALS neurons expressing mutant SOD1 and in neurons isolated from Alzheimer's models with mutant APP. In both cases, they observed a similar increase in Syntaphilin-positive MDVs and revival of mitochondrial motility.[109] This is an interesting model where MDV release selectively removed the machinery that anchors mitochondria in the terminal, allowing increased

Table 3. Disease conditions linked to MDVs

Disease	Molecular player	Functional process	MDV references
Neurodegeneration			
Parkinson's disease	PINK1, Parkin, VPS35	antigen presentation; quality control	McLelland et al.,[8] Braschi et al.,[10] Wang et al.,[17] Wang et al.,[18] Matheoud et al.,[19,24] and McLelland et al.[46]
Alzheimer's disease	SNPH	mitochondrial motility	Lin et al.[109]
Amyotrophic laterals sclerosis	SNPH	mitochondrial motility	Lin et al.[109]
Cardiac dysfunction			
Heart dysfunction	unknown	quality control; ROS signaling	Cadete et al.[110] and Li et al.[112]
Aging-related functional decline			
Stem cell function	SLC25A1	quality control; regulation of metabolism	Pouikli et al.[111]
Disturbed antimicrobial defense			
Phagosomal bacterial defense	Parkin	ROS signaling	Abuaita et al.[22]

mitochondrial motility and dynamics.[109] As mitochondria are transported back to the soma, they may recover or undergo mitophagy.

In the context of infection, MDV delivery to the late endosome led to the presentation of mitochondrial antigens.[19] This was repressed by PINK1 and Parkin, as outlined above. Importantly, *in vivo* studies infecting wild-type and PINK1−/− mice with *Citrobacter rodentium* showed a strong increase in mitochondrial antigen presentation from antigen-presenting cells within PINK1−/− mice.[24] This resulted in an expansion of cytotoxic $CD8^+$ T cells specific for mitochondrial antigens, demonstrating global effects on the peripheral immune system in genetically susceptible mice. Ultimately, these mice developed motor impairment, acutely reversible upon treatment with L-DOPA, highlighting a failure in dopaminergic neurons. Histological examination showed a retraction of dopaminergic neuron terminals from the striatum, consistent with the pathology.[24] Although more work is required to develop a detailed map of events from the initial gut infection to the peripheral immune system and loss of dopaminergic neuron terminals, it provided a new proof of concept linking MDV transport to the development of autoimmune pathways in a novel PD model.

MDVs were also examined in mouse heart and cultured H9c2 cardiomyocytes.[110,112] Although seen in steady state, MDVs rapidly increased ~2-fold upon treatment with doxorubicin, a cardiotoxic stress used as a chemotherapeutic in cancer patients.[110] Using high-resolution quantitative electron tomography in mouse heart, a series of both single- and double-membraned MDVs were observed, along with some autophagosomes containing larger mitochondria.[110] As in other cancer cell models, cultured differentiated H9c2 cells showed rapid induction of MDVs with doxorubicin, prior to any mitochondrial dysfunction or fragmentation, consistent with the emerging consensus that MDVs are a first-line defense to mitochondrial stress.[110] Another study exposed H9c2 cells to hypoxic stress to mimic I/R injury and again showed a rapid increase in MDVs peaking at 1 h, decreasing gradually.[112] Ultrastructural analysis of rat hearts after acute ischemia similarly revealed an increased generation of MDVs. At 12 h of hypoxia, H9c2 cells underwent apoptosis, prompting them to consider the initial generation of MDVs to be protective against death.[112] There are many questions remaining as to the fate and function of MDVs in the ischemic heart, but the data highlight the rapid generation of MDVs as a key early response to multiple mitochondrial stressors from cardiotoxins like doxorubicin to I/R.

Lastly, in a search for factors responsible for dysfunction in aging mesenchymal stem cells (MSC), the authors identified a highly selective, MDV-mediated removal of the mitochondrial citrate carrier CiC (SLC25A1). This carrier is required to export citrate, the precursor of acetyl-CoA, from the mitochondrial matrix to the cytosol. The loss of CiC resulted in the accumulation of citrate in mitochondria and the dramatic depletion of acetyl-CoA in the cytosol. This led to a block in chromatin acetylation that impaired osteogenesis.[111] Restoration of cytosolic levels of acetyl-CoA upon re-expression of CiC or exogenous addition of sodium acetate rescued the histone acetylation and osteogenic potential. This confirmed that the primary block in the supply of acetyl-CoA was at the level of the mitochondrial CiC. The loss was shown to be due to degradation of CiC in lysosomes and confocal imaging confirmed the presence of CiC in TOMM20+/PDH− MDVs. Ultrastructure imaging further confirmed the presence of double-membraned MDVs emerging from mitochondria in aged MSC. It remains unclear why or how the CiC is selected for degradation in aging MSC, but it has been established that there is increased mitochondrial stress with age, which may initiate the process.[111] Importantly this deficit was conserved in aged human MSC, offering new therapeutic potential for age-related osteoporosis by restoring the cytosolic pools of acetyl-CoA.[111] These findings highlight the capacity of MDV transport to sculpt the mitochondrial proteome and rewire metabolism.

CHALLENGES AND FUTURE DIRECTIONS

The discovery of MDVs marked a new understanding of the plasticity of mitochondrial membranes to sort cargoes into a variety of membrane-bound carriers with multiple fates and functions. Although initial definitions and mechanisms of MDVs continue to expand and evolve, the core principles have been established. Recent progress has been made in the identification of protein

machineries that guide their formation, which will inform future studies to better define the mechanisms of cargo selection. We now know that mitochondria also have the surprising capacity to generate very large, micron-sized compartments, which are highly selective in their cargo incorporation. Mitochondria are highly responsive to various cellular stressors and signaling cues, with rapid responses to generate vesicles in a variety of sizes and emanate tubular extensions that help to enrich cargoes. MDV transport is now a key aspect of mitochondrial biology as we continue to push forward our understanding of how they integrate complex behaviors as they shape-shift, fuse, sculpt, and contact other organelles to allow an incredible responsiveness to the metabolic needs of the cell. This review has also highlighted emerging pathophysiological implications of MDV transport in disease pathways, from neurodegenerative diseases to peroxisomal biogenesis disorders, heart disease, aging, cancer biology, innate and adaptive immunity, and more. It remains challenging to dissect the molecular mechanisms and cues that drive MDV release *in vivo*, and we await exciting new discoveries that will unravel the integration of MDVs with mitophagy, the links to the endomembrane system and peroxisome biology, and whatever unexpected surprises await.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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